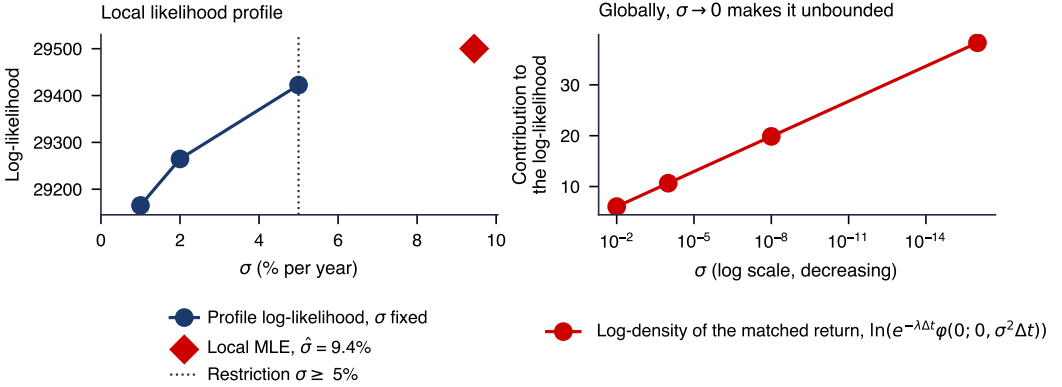
Local likelihood profile
Log-likelihood
29500
29400
29300
29200
0
2
4
6
8
10
σ (% per year)
Globally, σ → 0 makes it unbounded
Contribution to the log-likelihood
30
20
10
10^-2
10^-5
10^-8
10^-11
10^-14
σ (log scale, decreasing)
Profile log-likelihood, σ fixed
Local MLE, σ̂ = 9.4%
Restriction σ ≥ 5%
Log-density of the matched return, ln(e^-λΔt φ(0; 0, σ²Δt))